

Exact Diagonal and Mixed Reconstructed Correlator Decay from a Dobrushin Window for Finite Coupled $\mathbb{Z}_2$ Slices

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Abstract

We give one end-to-end account of a finite-volume chain that had previously been split between a Dobrushin comparison development and a reconstruction interface. For anisotropic coupled $\mathbb{Z}_2$ slices, the explicit condition

$$0 < \alpha < 1, \quad 2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha$$

first controls the intrinsic single-site interdependence matrix, then the finite-strip covariances, then the fluctuation norm of the normalized symmetric transfer kernel. An exact square-root dressing transports that bound to the reconstructed site form. Consequently, with $m = -\log \alpha > 0$, one obtains for every spatial extent and every finite temporal separation

$$|C_L^{\text{rec}}(u; n)| \leq \|u\|_2^2 e^{-mn}.$$

The exponent and prefactor are unchanged by reconstruction. This version also defines the mixed correlator in both abstract and reconstructed coordinates, proves their exact equality, and machine-checks

$$|B_L^{\text{rec}}(u, v; n)| \leq \|u\|_2 \|v\|_2 e^{-mn}$$

including the separate zero-time case. Here u and v range over the full real algebra of functions on a boundary slice. No sharper Dobrushin window, thermodynamic limit, continuum reconstruction, or Yang–Mills mass gap is claimed. The analytic estimate is inherited; the new formal increment is the mixed observable endpoint and its inclusion in the uniform theorem.

1 Result and exact scope

Let

$$X_L = (\text{Fin}(L+1) \rightarrow \text{Fin } 2), \quad \mathcal{A}_L = \mathbb{R}^{X_L}.$$

The space $\mathcal{A}_L$ is not merely a chosen vector class: with pointwise multiplication it is the complete finite-dimensional commutative algebra of real observables on one boundary slice. Thus every cylinder observable that depends on that whole slice, as well as every finite sum and pointwise product of such observables, is represented by some $u \in \mathcal{A}_L$. What remains finite is the slice itself and the temporal separation.

The slice weight $w_{\gamma, L} : X_L \rightarrow \mathbb{R}_{>0}$ contains the spatial coupling γ . The temporal kernel K_β is dressed symmetrically as

$$S_{w, \beta} = \text{diag}(\sqrt{w}) K_\beta \text{diag}(\sqrt{w}).$$

For positive Perron data (λ_L, Ω_L) put

$$P_L = \lambda_L^{-1} S_{w_{\gamma, L}, \beta}, \quad P_L \Omega_L = \Omega_L, \quad \|\Omega_L\|_2 = 1.$$

The reconstructed site form and forced transfer operator are denoted by $\mathbf{q}_w$ and $T_{w, \beta}$. Their precise finite sums appear in Section 5.

Theorem 1.1 (end-to-end finite reconstructed clustering). *Assume*

$$0 < \alpha < 1, \quad 2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha.$$

Set $m = -\log \alpha$. Then $m > 0$ and, for every $L \geq 0$, there are positive normalized Perron data (λ_L, Ω_L) such that

$$\|P_L - |\Omega_L\rangle\langle\Omega_L|\| \leq e^{-m}$$

and, for all $u \in \mathcal{A}_L$ and $n \in \mathbb{N}$,

$$|C_L^{\text{rec}}(u; n)| \leq \|u\|_2^2 e^{-mn}.$$

For all $u, v \in \mathcal{A}_L$ and $n \in \mathbb{N}$ one also has

$$|B_L^{\text{rec}}(u, v; n)| \leq \|u\|_2 \|v\|_2 e^{-mn}.$$

Moreover $C_L^{\text{rec}}(u; n)$ is exactly the Euclidean connected correlator of P_L , and $B_L^{\text{rec}}(u, v; n)$ is exactly its mixed connected matrix element, not merely bounded by it.

The binder order is load-bearing:

$$\exists m > 0 \forall L \exists (\lambda_L, \Omega_L) \forall u, v, n.$$

Perron data may depend on the volume; the exponent does not. The theorem is the mathematical statement exposed by `os_reconstruction_uniform_gap`. The following sections give the entire implication chain rather than referring to the analytic input as a black box.

2 The intrinsic Dobrushin estimate

Let I be a finite site set, $S = \{\pm 1\}$, and let $W : S^I \rightarrow \mathbb{R}_{>0}$ be a strictly positive Gibbs weight. Its normalized measure and heat-bath conditional are

$$\mu_W(\eta) = \frac{W(\eta)}{\sum_{\xi} W(\xi)}, \quad p_i^W(\eta, s) = \frac{W(\eta^{i \rightarrow s})}{\sum_{t \in S} W(\eta^{i \rightarrow t})}.$$

Define the intrinsic interdependence coefficient

$$c_{ik}^W = \max_{\eta \equiv \eta' \text{ off } k} \|p_i^W(\eta, \cdot) - p_i^W(\eta', \cdot)\|_{\text{TV}}.$$

For an observable f let

$$\delta_i(f) = \max_{\eta \equiv \eta' \text{ off } i} |f(\eta) - f(\eta')|.$$

Lemma 2.1 (finite Dobrushin comparison). *If $\sum_k c_{ik}^W \leq \alpha < 1$ for every i , then*

$$|\text{Cov}_{\mu_W}(f, g)| \leq \frac{1}{4} \sum_{i,j} \delta_i(f) \left(\sum_{q \geq 0} (c_{ij}^W)^q \right) \delta_j(g).$$

If $c_{ik}^W = 0$ whenever $d(i, k) > 1$, then

$$|\text{Cov}_{\mu_W}(f, g)| \leq \frac{1}{4(1-\alpha)} \sum_{i,j} \delta_i(f) \delta_j(g) \alpha^{d(i,j)}.$$

Proof. The heat-bath update E_i is invariant under μ_W . Indeed, after expanding $\mathbb{E}_{\mu_W}[E_i F]$ over (η, s) , the involution $(\eta, s) \mapsto (\eta^{i \rightarrow s}, \eta_i)$ preserves the local partition sum and cancels its denominator. The conditional Grüss estimate contributes the factor $1/4$. A single update transports the oscillation vector by c^W ; iteration and invariance give the Neumann resolvent $D = \sum_{q \geq 0} (c^W)^q$. The row-sum hypothesis makes the series entrywise summable. Under range one, $(c^W)_{ij}^q = 0$ for $q < d(i, j)$ and its row sum is at most α^q , hence

$$D_{ij} \leq \sum_{q \geq d(i,j)} \alpha^q = \frac{\alpha^{d(i,j)}}{1 - \alpha}.$$

Substitution proves both bounds. The corresponding checked endpoints are `gibbs_covar_le_resolvent` and `gibbs_covar_exp_decay`. $\square$

3 The anisotropic rectangle and the common rate

For a symmetric zero-diagonal Ising coupling J , write

$$W_J(\eta) = \exp\left(\frac{1}{2} \sum_{a,b} J_{ab} \eta_a \eta_b\right).$$

The heat-bath conditional is the sigmoid of the local field $h_i(\eta) = \sum_k J_{ik} \eta_k$:

$$p_i^{W_J}(\eta, +1) = \frac{1 + \tanh h_i(\eta)}{2}.$$

Changing spin k moves h_i by at most $2|J_{ik}|$. The elementary two-point sigmoid calculation therefore gives

$$c_{ik}^{W_J} \leq \tanh |J_{ik}|.$$

This is an envelope, not a claim that the intrinsic coefficient always attains it.

Consider a finite rectangular strip, with coupling β in the temporal direction and γ in the spatial direction, and free boundary. Every site has at most two neighbors of each type, hence

$$\sum_k c_{ik}^{W_J} \leq 2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha.$$

Boundary sites have a smaller row sum. Since the coupling has range one for Manhattan distance, Lemma 2.1 yields, uniformly in both side lengths,

$$|\text{Cov}_{\mu_{L,T}}(f, g)| \leq \frac{1}{4(1 - \alpha)} \sum_{p,q} \delta_p(f) \delta_q(g) \alpha^{d_{\square}(p,q)}.$$

For observables supported on the two end slices of a strip of temporal length n , every contributing pair has $d_{\square}(p, q) \geq n$. Therefore

$$|\text{Cov}_{\mu_{L,n}}(f, g)| \leq C_{L,f,g} \alpha^n, \tag{1}$$

where $C_{L,f,g}$ may depend on the spatial extent and the observables, but not on n . This precise freedom is sufficient for the operator transport. The checked rectangle endpoints are `rect_ising_uniform_decay` and `rect_ising_uniform_two_point`.

4 From strip covariances to an operator gap

Let M be a symmetric matrix on a finite set X , with a strictly positive vector Ω satisfying $M\Omega = \Omega$ and $\|\Omega\|_2 = 1$. For a path $v = (v_0, \dots, v_n)$ define

$$W_n(v) = \Omega(v_0) \prod_{k < n} M_{v_k v_{k+1}} \Omega(v_n).$$

Successively summing the eigen-relation shows that $\sum_v W_n(v) = 1$. For endpoint observables $f, g : X \rightarrow \mathbb{R}$, direct matrix multiplication gives

$$\text{bandCov}_n(f, g) = \langle (f - \mathbb{E}f)\Omega, M^n(g - \mathbb{E}g)\Omega \rangle. \quad (2)$$

No spectral positivity is used; symmetry and the Perron relation suffice.

For the physical slice weight w and temporal kernel K_β , take

$$S = \text{diag}(\sqrt{w})K_\beta \text{diag}(\sqrt{w}), \quad M = S/\lambda, \quad \psi = \Omega/\sqrt{w}.$$

Then the band weight is exactly a free-strip Gibbs weight with the two endpoint factors ψ , divided by λ^n . The identity is pointwise, so the boundary factors change only $C_{L,f,g}$ in (1); they do not change the common rate α . The currying bijection between rectangle configurations and slice paths then identifies the free strip covariance with the rectangle covariance.

Proposition 4.1 (uniform projected norm). *Under the window of Theorem 1.1, for every L the normalized symmetric slice kernel has Perron data with*

$$\|M_L - |\Omega_L\rangle\langle\Omega_L|\| \leq \alpha.$$

Proof. Equation (1), the exact tilt identity, and (2) give, for every vector v , a constant $C_{L,v}$ with

$$|\langle v, (M_L^n - |\Omega_L\rangle\langle\Omega_L|)v \rangle| \leq C_{L,v}\alpha^n.$$

The fluctuation operator is self-adjoint. If its norm exceeded α , a finite-dimensional spectral vector in the fluctuation sector would make the left side decay at a strictly slower exponential rate, contradicting the display. Hence its norm is at most $\alpha = e^{-m}$ with $m = -\log \alpha$. In Lean this is the composition ending at `dobrushin_ising_uniform_gap`, through `abstract_uniform_gap`. $\square$

This step explains why a volume-dependent prefactor in the measure estimate does not spoil the conclusion: the contradiction is performed separately in each finite volume, whereas the forbidden rate is common.

5 Exact reconstruction

For $u, v \in \mathcal{A}_L$ define the reconstructed site form

$$\mathfrak{q}_w(Q_w u, Q_w v) = \sum_{\sigma \in X_L} u(\sigma)v(\sigma), \quad (Q_w u)(\sigma) = \sqrt{w(\sigma)}u(\sigma).$$

In the implementation the form is first written with the weight w and the inverse square-root dressing; strict positivity of w cancels those factors exactly. Consequently Q_w is an isometry from the Euclidean boundary space onto the real reconstructed site coordinates.

The forced transfer matrix satisfies the exact intertwining relation

$$T_{w,\beta}Q_w u = Q_w S_{w,\beta} u. \quad (3)$$

Induction gives the equality for every finite iterate. After Perron normalization,

$$(\lambda^{-1}T_{w,\beta})^n Q_w u = Q_w P^n u.$$

The checked declarations are `siteForm_qEmbed`, `transferOp_qEmbed`, `transferOp_iterate_qEmbed`, and `transferOp_qEmbed_tilt_iterate`.

Define the reconstructed connected autocorrelator by

$$C_{w,\beta,\lambda,\Omega}^{\text{rec}}(u; n) = \text{Re } \mathfrak{q}_w \left(Q_w u, (\lambda^{-1}T_{w,\beta})^n Q_w u \right) - \text{Re } \mathfrak{q}_w(Q_w \Omega, Q_w \Omega)^2.$$

Equations above give the exact identity

$$C_{w,\beta,\lambda,\Omega}^{\text{rec}}(u; n) = \langle u, P^n u \rangle - \langle \Omega, u \rangle^2. \quad (4)$$

Thus, writing $\Pi = |\Omega\rangle\langle\Omega|$ and $R_n = P^n - \Pi$,

$$|C^{\text{rec}}(u; n)| = |\langle u, R_n u \rangle| \leq \|u\|_2^2 \|P - \Pi\|^n.$$

For $n \geq 1$ this uses $P^n - \Pi = (P - \Pi)^n$; for $n = 0$ it uses $\|I - \Pi\| \leq 1$. This separate zero-time case is necessary because the zeroth power of $P - \Pi$ is I , not $I - \Pi$. Combining this with Proposition 4.1 proves Theorem 1.1. The endpoint is checked as `reconstructedConnCorr_eq_connCorr` and `reconstructedConnCorr_decay`.

6 The full boundary observable algebra

The diagonal notation in (4) can obscure the scope. For $u, v \in \mathcal{A}_L$ define the mixed connected form

$$B_L^{\text{rec}}(u, v; n) = \langle u, P_L^n v \rangle - \langle \Omega_L, u \rangle \langle \Omega_L, v \rangle.$$

It is bilinear and, since P_L is symmetric, symmetric in u, v . The implementation now contains the abstract definition `mixedConnCorr` and the reconstructed definition `reconstructedMixedConnCorr`; the latter is written directly with the two reconstructed boundary vectors rather than obtained by a paper-only polarization step.

Corollary 6.1 (mixed boundary observables). *Under the hypotheses of Theorem 1.1,*

$$|B_L^{\text{rec}}(u, v; n)| \leq \|u\|_2 \|v\|_2 e^{-mn}$$

for all $u, v \in \mathcal{A}_L$ and $n \in \mathbb{N}$.

Proof. Put $\Pi_L = |\Omega_L\rangle\langle\Omega_L|$. For $n \geq 1$, the vacuum projection commutes with P_L and $P_L^n - \Pi_L = (P_L - \Pi_L)^n$. Cauchy-Schwarz and $\|P_L - \Pi_L\| \leq e^{-m}$ give the claim. At $n = 0$, use $\|I - \Pi_L\| \leq 1 = e^{-m \cdot 0}$. $\square$

The positive-time identity is checked by `VacuumTransfer.mixedConnCorr_eq`. The zero-time contraction $\|I - \Pi_L\| \leq 1$ is now a named theorem, `norm_one_sub_vacuumProjection_le`; both cases are assembled by `mixed_clustering_of_gap`. Exact transport to reconstructed coordinates is `reconstructedMixedConnCorr_eq_mixedConnCorr`, and the product-norm estimate is `reconstructedMixedConnCorr_decay`. Finally, `os_reconstruction_uniform_gap` exports the mixed clause under the same volume-independent exponent as the diagonal clause.

This formal endpoint shows that the result is not restricted to one preferred scalar observable. Every real boundary function is admitted, and any finite covariance matrix built from a family $u_1, \dots, u_k \in \mathcal{A}_L$ inherits the same entrywise exponential rate. It does not enlarge the support beyond one finite slice or construct an infinite-time observable algebra.

7 Machine-checked map and reproducibility

This version changes two Lean modules. Relative to v3 it adds two definitions and five named theorems: the abstract and reconstructed mixed correlators, the positive-time structural identity, the zero-time projection contraction, the mixed gap-to-clustering estimate, exact reconstruction, and reconstructed mixed decay. It also strengthens the uniform endpoint by adding the mixed clause without changing its Dobrushin hypotheses.

The exact reconstruction source has 11621 LF bytes and SHA-256

a3089a961a3f7659c9a5317981168ffcccd980bffc10f894adab1788076ed43b

and its materialized .olean has SHA-256

739a3c64a53802e1f6483ce63b53f72b1ce06cdbf0ddd856b6efbb70f7f5ffa7

The abstract bridge source has 36484 LF bytes and SHA-256

1dd660dceb92b133a5dd75e4e9a7bb3afad9d5d635010c61b8586c686227cbe9

and its materialized .olean has SHA-256

77eb1ed915efc4c3ce7905de7b72bc3d82e117d72d18a8bab21df2da35da9555

Validation used Colab Pro+ with CPU and high RAM; no local Lean or Lake command was run. The target build completed 8191 jobs with exit code zero. The dependency build completed 8482 jobs with exit code zero. Four exhaustive oracle chunks were derived from the exact local `oracle_check.lean` body and emitted 3054 reports: 3027 used only Lean’s standard axioms `Classical.choice`, `Quot.sound`, and `propext`; 27 were axiom-free; there were zero nonstandard axioms, zero `sorryAx`, and zero core errors. The runtime was then disconnected and deleted. A frozen-object manifest records the body commit, LF/CRLF projections, build logs, oracle hashes, and PDF hashes for independent assessment.

Mathematical step	Checked endpoint or source lane
Intrinsic heat-bath comparison	<code>gibbs_covar_le_resolvent</code>
Range-one exponential decay	<code>gibbs_covar_exp_decay</code>
Anisotropic rectangle, all volumes	<code>rect_ising_uniform_decay</code>
Band covariance identity	<code>band_covariance_eq</code>
Measure decay to operator norm	<code>abstract_uniform_gap</code>
Perron tilt and rectangle feed	<code>dobrushin_ising_uniform_gap</code>
Exact reconstructed correlator	<code>reconstructedConnCorr_eq_connCorr</code>
Reconstructed decay	<code>reconstructedConnCorr_decay</code>
Exact mixed reconstructed correlator	<code>reconstructedMixedConnCorr_eq_mixedConnCorr</code>
Mixed reconstructed decay	<code>reconstructedMixedConnCorr_decay</code>
Uniform physical endpoint	<code>os_reconstruction_uniform_gap</code>

The analytic declarations in the first six rows predate the reconstruction module. This paper does not relabel them as new; it reproduces their argument so the endpoint can be read without consulting a separate paper.

8 Limitations and precise contribution

The proof is finite-volume throughout. It does not construct an infinite-volume Gibbs state, prove boundary-condition independence, take a thermodynamic or continuum limit, reconstruct Wightman fields, or prove a Yang–Mills mass gap. The Dobrushin window is sufficient and

deliberately conservative; for the two-dimensional Ising cell it is weaker than the exact solution. No improvement of that window or of $m = -\log \alpha$ is claimed.

What the integrated statement adds is exact compositional control. The classical single-site condition is carried through a finite rectangle, a Perron boundary tilt, a projected operator norm, and the reconstructed site form. Every transition is an equality or a bound with its quantifiers visible, and reconstruction loses neither rate nor prefactor. The mixed-observable theorem makes explicit—in the formal interface, not only in prose—that two independent boundary variables range over the full real function algebra of the finite slice. This is a genuine formal increment but no new analytic estimate: it packages the inherited operator-norm gap through exact reconstruction with the sharp product prefactor. These are useful endpoints of a formalization program, but they remain finite and high-temperature.

Primary references

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